Multi-Component Air Differential Detection and Graded Purification Device

Technical Field

The present invention relates to the technical field of air quality monitoring and air purification, and specifically relates to a multi-component air differential detection and graded purification device.

Background Technology

With the development of modern industry and the complication of indoor environments, air pollutants exhibit multi-source and complex characteristics, mainly including suspended fine particulate matter (PM2.5), acidic harmful gases (such as sulfur dioxide SO2), and highly lethal reducing toxic gases (such as carbon monoxide CO).

Existing portable or small-scale air purification devices generally have the following defects:

1. Single purification mechanism: Most purifiers solely rely on a single HEPA filter or passive activated carbon adsorption. They cannot achieve efficient decomposition of trace amounts of highly toxic carbon monoxide at room temperature. The adsorption capacity for acidic gases is limited, and secondary pollution caused by saturation desorption is prone to occur.

2. Lack of closed-loop efficacy evaluation: Traditional equipment mostly sets a single detection sensor at the air inlet or on the surface of the casing, unable to monitor the air quality at the air outlet in real time, and unable to objectively

evaluate the real-time purification efficacy of the device itself, leading to inaccurate early warnings when filter elements and reaction reagents fail.

3. Lagging control strategy: Existing devices mostly use fixed steady-state threshold judgments. Their response is sluggish during sudden high-concentration pollution indoors (such as gas leaks or the generation of a large amount of smoke in a short time), and they cannot issue proactive safety warnings in a timely manner at the initial stage of a concentration surge.

Summary of the Invention

To overcome the deficiencies of the prior art, the present invention provides a multi-component air differential detection and graded purification device, which can achieve graded, targeted, and efficient purification of particulate matter, acidic gases, and carbon monoxide, and can compare concentrations before and after purification in real time to evaluate purification efficacy and provide rapid early warnings.

The technical solutions adopted by the present invention are as follows:

A multi-component air differential detection and graded purification device comprises a detection box (1) and a base (2) disposed below the detection box (1). The side walls of the detection box (1) are respectively penetrated to define an air inlet channel (3) and an air outlet channel (6). The inside of the detection box (1) defines an S-shaped folded flow channel (13) communicating the air inlet channel (3) and the air outlet channel (6).

The S-shaped folded flow channel (13) is sequentially provided with an intake fan (10), an adjustable Tesla speed-limiting valve (19), a front detection chamber (11), a high-temperature catalytic reaction unit (16), an alkaline chemical adsorption unit

(17), and a rear detection chamber (18) along the airflow direction from the air inlet channel (3) to the air outlet channel (6).

The adjustable Tesla speed–limiting valve (19) comprises upper and lower variable pipe walls hinged via flow channel rotating shafts (192), and variable baffles (191) are provided at branch intersections inside the adjustable Tesla speed–limiting valve (19).

Detection units (12) are installed inside both the front detection chamber (11) and the rear detection chamber (18). The detection unit (12) includes a baseboard (121) and a PM2.5 laser scattering sensor (122), a sulfur dioxide electrochemical sensor (123), and a carbon monoxide sensor (124) integrated on the baseboard (121).

The high–temperature catalytic reaction unit (16) comprises a high–temperature resistant heating chamber (161), ACO–2 catalyst filter elements (162) horizontally embedded in the high–temperature resistant heating chamber (161), an atomizing nozzle (163) disposed at the upper inner part of the high–temperature resistant heating chamber (161), a liquid storage tank (164) installed on the outer side of the high–temperature resistant heating chamber (161), and a micro pump (166) communicating with the atomizing nozzle (163) and the liquid storage tank (164). The top of the liquid storage tank (164) is connected with a hydrogen peroxide replenishing pipe (5) penetrating out of the top of the detection box (1).

The alkaline chemical adsorption unit (17) comprises an adsorption chamber (171) and sodium hydroxide modified activated carbon porous granular filter elements (172) plugged into the adsorption chamber (171).

A main control circuit board (14) and a storage battery (15) are installed inside the base (2). The intake fan (10), the detection units (12), the micro pump (166), and the storage battery (15) are all electrically connected to the main control circuit board (14).

Further, a buzzer (4) electrically connected to the main control circuit board (14) is installed at the top of the detection box (1). A display screen (7), LED indicator lights (8), and a charging interface (9) are installed on the front panel of the base (2). Both the display screen (7) and the LED indicator lights (8) are electrically connected to the main control circuit board (14), and the charging interface (9) is electrically connected to the storage battery (15).

Further, when the detection value of the carbon monoxide sensor (124) in the front detection chamber (11) exceeds an alarm purification threshold, the main control circuit board (14) controls the buzzer (4) and the LED indicator lights (8) to issue an alarm signal, starts the intake fan (10) and the micro pump (166), and detects the exhaust concentration through the carbon monoxide sensor (124) in the rear detection chamber (18) and outputs it via the display screen (7).

Further, when the detection value of the PM2.5 laser scattering sensor (122) or the sulfur dioxide electrochemical sensor (123) in the front detection chamber (11) exceeds an alarm purification threshold, the main control circuit board (14) controls the buzzer (4) and the LED indicator lights (8) to issue an alarm signal, starts the intake fan (10), and detects the exhaust concentration through the PM2.5 laser scattering sensor (122) and the sulfur dioxide electrochemical sensor (123) in the rear detection chamber (18) and outputs it via the display screen (7).

Further, the main control circuit board (14) calculates the concentration rise rate of each pollutant at a fixed sampling cycle; when the concentration rise rate of any component within a preset time window exceeds a sudden-increase rate threshold and the real-time concentration reaches a warning threshold, the main control circuit board (14) proactively controls the buzzer (4) to issue an early warning signal.

Further, a liquid level sensor (165) electrically connected to the main control circuit board (14) is installed inside the liquid storage tank (164).

Further, the sodium hydroxide modified activated carbon porous granular filter elements (172) are provided in multiple layers at intervals along the vertical direction of the adsorption chamber (171), and each layer of the sodium hydroxide modified activated carbon porous granular filter element (172) is hermetically plugged into the adsorption chamber (171) via drawer-type slide rails.

The beneficial effects of the present invention are:

1. Multi-stage targeted purification: Through the dual reaction paths of "high-temperature catalytic oxidation (decomposing CO) + alkaline chemical neutralization and adsorption (removing SO2 and particulate matter)", chamber-specific and highly efficient removal of pollutants of different properties is achieved;

2. Differential closed-loop efficacy monitoring: The front and rear groups of detection chambers form a differential comparison system, which can visually output the exhaust concentration after purification and monitor the consumption life of the filter elements and reagents in real time;

3. Structural integration and quick-release maintenance: The reaction chamber and adsorption filter elements adopt a drawer-type slide rail and pipe connection layout, facilitating the independent replacement of consumables. Paired with an independent bottom circuit and power supply bin, it has strong anti-interference capabilities;

4. Dynamic sudden-increase rapid response and adaptive speed regulation: Proactive early warnings of sudden pollution are achieved through the concentration differential slope; by using the adjustable Tesla speed-limiting valve to dynamically change the cross-sectional area and baffle flow resistance, adaptive extension of the airflow residence time is achieved without reducing the total flow rate, significantly enhancing the response agility and purification depth for extremely dangerous working conditions.

Brief Description of Drawings

Figure 1 is a schematic diagram of the overall external three-dimensional structure of an embodiment of the present invention.

Figure 2 is a schematic diagram of the internal structural cross-section and flow channel of an embodiment of the present invention.

Figure 3 is a schematic diagram of the three-dimensional structure of the high-temperature catalytic reaction unit in an embodiment of the present invention.

Figure 4 is a schematic diagram of the three-dimensional structure of the alkaline chemical adsorption unit in an embodiment of the present invention.

Figure 5 is a schematic diagram of the structural layout of the detection unit in an embodiment of the present invention.

Figure 6 is a schematic diagram of the internal structure and variable baffle control principle of the adjustable Tesla speed-limiting valve in an embodiment of the present invention.

In the figures: 1-Detection box, 2-Base, 3-Air inlet channel, 4-Buzzer, 5-Hydrogen peroxide replenishing pipe, 6-Air outlet channel, 7-Display screen, 8-LED indicator lights, 9-Charging interface, 10-Intake fan, 11-Front detection chamber, 12-Detection unit, 121-Baseboard, 122-PM2.5 laser scattering sensor, 123-Sulfur dioxide electrochemical sensor, 124-Carbon monoxide sensor, 13-S-shaped folded flow channel, 14-Main control circuit board, 15-Storage battery, 16-High-temperature catalytic reaction unit, 161-High-temperature resistant heating chamber, 162-ACO-2 catalyst filter element, 163-Atomizing nozzle, 164-Liquid storage tank, 165-Liquid level sensor, 166-Micro pump, 17-Alkaline chemical adsorption unit, 171-Adsorption chamber, 172-Sodium hydroxide modified activated

carbon porous granular filter element, 18–Rear detection chamber, 19–Adjustable Tesla speed–limiting valve, 191–Variable baffle, 192–Flow channel rotating shaft.

Detailed Description of Embodiments

The present invention will be further described in detail below in conjunction with the drawings and specific embodiments.

As shown in Figures 1 and 2, a multi–component air differential detection and graded purification device provided in this embodiment comprises an upper detection box (1) and a base (2) disposed below the detection box (1). An air inlet channel (3) is provided at the upper left side of the detection box (1), and an air outlet channel (6) is provided at the upper right side. An S–shaped folded flow channel (13) is constructed inside the detection box (1) by partition plates. The S–shaped folded flow channel (13) communicates the air inlet channel (3) and the air outlet channel (6), effectively extending the flow path and reaction residence time of the gas inside the box body.

Along the airflow direction, the S–shaped folded flow channel (13) is sequentially equipped with an intake fan (10), an adjustable Tesla speed–limiting valve (19), a front detection chamber (11), a high–temperature catalytic reaction unit (16), an alkaline chemical adsorption unit (17), and a rear detection chamber (18). The intake fan (10) is located in the downstream pipeline of the air inlet channel (3) and employs a DC PWM speed–regulated fan.

As shown in Figure 6, the adjustable Tesla speed–limiting valve (19) is disposed immediately downstream of the intake fan (10). The upper and lower pipe walls of the adjustable Tesla speed–limiting valve (19) are hinged via flow channel rotating shafts (192) provided at the inlet end, thereby allowing the angle between the pipe walls to be changed. According to the fluid continuity equation $Q = v \times A$, the main control circuit board (14) dynamically controls and adjusts the angle between the

upper and lower pipe walls of the adjustable Tesla speed–limiting valve (19) based on the detection data of the front detection chamber (11) and then according to a preset gas flow velocity required for sufficient reaction, to alter the cross–sectional area A. On the premise of not reducing the total flow rate Q, the effect of slowing down the flow velocity v is achieved by increasing the cross–sectional area A. At the same time, variable baffles (191) are added at each branch intersection inside the adjustable Tesla speed–limiting valve (19). By dynamically adjusting the deflection angle of the variable baffles (191), the main control circuit board (14) changes the area of the intersections, thereby altering the flow velocity and flow resistance, ensuring that harmful gases have adequate reaction residence time in the subsequent units.

As shown in Figures 2 and 5, the front detection chamber (11) and the rear detection chamber (18) are respectively disposed in straight pipe sections at both ends of the S–shaped folded flow channel (13), and structurally identical detection units (12) are installed inside them. The detection unit (12) includes a baseboard (121), and a PM2.5 laser scattering sensor (122), a sulfur dioxide electrochemical sensor (123), and a carbon monoxide sensor (124) are integrated and fixed on the baseboard (121). The front detection chamber (11) is used to collect the original pollutant gas concentration entering the device, and the rear detection chamber (18) is used to collect the exhaust concentration after purification.

As shown in Figures 2 and 3, the high–temperature catalytic reaction unit (16) includes a high–temperature resistant heating chamber (161). Heating tubes are pre–buried inside the high–temperature resistant heating chamber (161), and two layers of ACO–2 catalyst filter elements (162) are horizontally embedded in the high–temperature resistant heating chamber (161). An atomizing nozzle (163) is erected at the upper inner part of the high–temperature resistant heating chamber (161). A liquid storage tank (164) is fixed on the outer side wall of the high–

temperature resistant heating chamber (161). A micro pump (166) is installed on the top of the liquid storage tank (164), and the output end of the micro pump (166) communicates with the atomizing nozzle (163) via a pipeline. A liquid level sensor (165) is provided inside the liquid storage tank (164). The top of the liquid storage tank (164) is also drawn upwards with a hydrogen peroxide replenishing pipe (5), which penetrates upwards out of the outside of the top plate of the detection box (1), facilitating direct external liquid injection. When gas containing CO passes through, the micro pump (166) extracts hydrogen peroxide and sprays it through the atomizing nozzle (163), profoundly oxidizing CO into carbon dioxide under the synergy of heating and the ACO-2 catalyst.

As shown in Figures 2 and 4, the alkaline chemical adsorption unit (17) is located downstream of the high-temperature catalytic reaction unit (16) and includes an adsorption chamber (171). Three layers of sodium hydroxide modified activated carbon porous granular filter elements (172) are slidably plugged into the inside of the adsorption chamber (171) at equal intervals along the vertical direction. Each layer of the filter element is snapped into the adsorption chamber (171) via drawer-type slide rails. On one hand, the porous media of the activated carbon is utilized to intercept and filter fine particulate matter (PM2.5); on the other hand, the strongly alkaline sodium hydroxide loaded via modification is utilized to undergo chemical neutralization and absorption with acidic gas SO2, generating stable solid salts to achieve solidified removal.

As shown in Figures 1 and 2, the base (2) internally integrates control and power systems, including a main control circuit board (14) and a storage battery (15). A buzzer (4) is installed at the top of the detection box (1), and a display screen (7), several LED indicator lights (8), and a charging interface (9) are embedded on the beveled front panel of the base (2). The main control circuit board (14) is electrically connected to the intake fan (10), detection units (12), micro pump (166), liquid level

sensor (165), buzzer (4), display screen (7), LED indicator lights (8), and storage battery (15) respectively. The charging interface (9) is electrically connected to the storage battery (15) to achieve portable power supply.

The specific working and regulation processes of this device are as follows:

1. Carbon monoxide pollution response process: When the reading of the carbon monoxide sensor (124) in the front detection chamber (11) reaches or exceeds a preset alarm purification threshold, the main control circuit board (14) drives the buzzer (4) and LED indicator lights (8) to alarm, starts the intake fan (10) for air intake, simultaneously turns on the micro pump (166) to spray hydrogen peroxide and conduct heating catalytic decomposition inside the high-temperature catalytic reaction unit (16), concurrently catalyzing the oxidation of carbon monoxide; the main control circuit board (14) continuously collects readings from the carbon monoxide sensor (124) in the rear detection chamber (18) and displays the outlet concentration and purification efficiency in real time on the display screen (7).

2. PM2.5 and sulfur dioxide pollution response process: When the PM2.5 laser scattering sensor (122) or sulfur dioxide electrochemical sensor (123) in the front detection chamber (11) exceeds the standard, the main control circuit board (14) issues an alarm and drives the intake fan (10) to run. The dust-laden and acidic gases directly pass through the alkaline chemical adsorption unit (17) to complete particulate adsorption and chemical neutralization, at which time the micro pump (166) remains in standby for energy saving. The rear detection chamber (18) detects exhaust PM2.5 and SO2 content in real time and refreshes it on the display screen (7).

3. Sudden-increase rapid early warning process: The main control circuit board (14) monitors the change slope of the concentration of each component in real time. When it detects that the concentration rise rate of any component exceeds a

sudden–increase rate threshold and the real–time concentration reaches a warning threshold, the main control circuit board (14) proactively drives the buzzer (4) to issue a rapid early warning signal, achieving proactive defense against fire hazards or severe leaks.